DATA ANALYTICS CASE STUDY · LAGOS FASHION

Which social platform makes Lagos fashion customers buy?

I audited 4 Lagos fashion brands' Instagram to find out. The answer surprised me.

0 of 280

recent comments asked about price, ordering or delivery.

4	80	553
competitor brands	posts scraped	comments classified with NLP

Swipe → what the data could and couldn't answer

THE BRIEF → WHAT I BUILT

“Audit 5 competitors across platforms to decide the ad budget.”

Based on a real Upwork job. Instagram rate-limited my own scraper, so I collected the public data with Apify.

1. Scraping · Playwright attempt + Apify export, 80 posts & 553 comments
2. Python ETL · validation, deduplication, provenance hashes, UTC→Lagos time
3. NLP intent classifier · negation, word boundaries, Nigerian Pidgin (“Wetin be the price?”)
4. SQL warehouse · SQLite/PostgreSQL, idempotent loads, SQL ⇄ Python checks
5. Streamlit dashboard + report · 125 automated tests

FINDING 1 · THE PURCHASE-INTENT FUNNEL

Comments are for applause, not checkout.

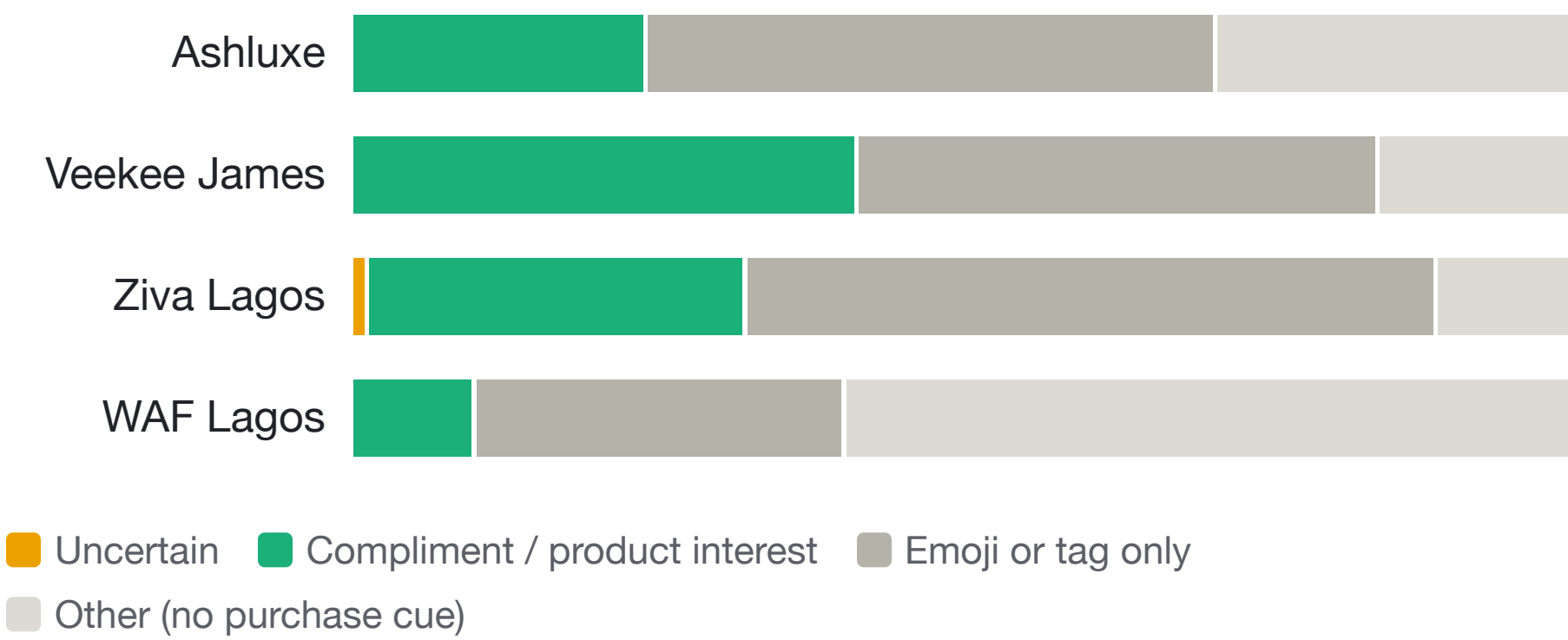
Last 60 days of public comments on 4 brands' posts:



Buying conversations most likely move to **DMs and WhatsApp**, which public data can't see. Across *all* collected posts, only 8 of 553 comments (1.4%) showed any shopping signal.

FINDING 2 · WHAT AUDIENCES ACTUALLY WRITE

47% emoji or tags only. 30% compliments.



65%

of Ziva Lagos' captions show a ₦ price

0%

is the highest share for any other brand: a clear difference in selling style

Seller behaviour differs a lot. Audience behaviour barely does: praise everywhere, buying questions almost nowhere.

FINDING 3 · THE HONEST ANSWER

“Which platform should get the budget?”

Not answerable from this data.

Only Instagram could be collected. Any “50% Instagram / 30% TikTok” split would have been invented. My first version of this project did exactly that — auditing it was the most valuable part.

What I recommended instead: a 6-week attribution test

Wk 1	Separate WhatsApp link, UTM-tagged link and discount code per platform
Wk 2–5	Equal small spend on Instagram, TikTok, WhatsApp Status; A/B test prices in captions
Wk 6	Set the budget split from **cost per order**, not likes

WHAT THIS PROJECT DEMONSTRATES

From messy scraped data to a decision a business can trust.

Web scraping · Python · pandas · ETL & data quality · NLP / text classification · SQL · PostgreSQL · Statistics · confidence intervals · Sensitivity analysis · Streamlit · Altair · pytest · 125 tests · Data storytelling

80
posts processed with full provenance

553
comments classified (5 intent levels)

125
automated tests passing

Code, dashboard & technical audit

github.com/Shieldemma/fashion-competitor-analytics